

Francesco **Baiocchi**

AI Engineer · MSc Student in AI

Rieti, Italy

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Summary

AI engineer and MSc student in Artificial Intelligence (University of Bologna) with industry experience in **Document AI** at LetXbe AI (Paris). I work end to end on deep learning systems — data pipelines, training, evaluation and interpretability — across **Vision-Language Models**, **LLM fine-tuning** (QLoRA), **semantic segmentation** and **graph neural networks**, in PyTorch and Hugging Face.

Education

Alma Mater Studiorum - University of Bologna

Bologna, Italy

MSc in Artificial Intelligence — *Sep. 2023 - Present*

- **Industrial thesis with LetXbe AI**: Vision-Language Models for Document Understanding (DocVQA, visual grounding, interpretability).
- **Coursework**: Deep Learning, Machine Learning for Computer Vision, NLP, Machine Learning, Data Mining, Combinatorial Optimization.
- **Stack**: Python, PyTorch, PyTorch Geometric, Hugging Face (Transformers, PEFT), scikit-learn, OpenCV, Weights & Biases, Quarto.
- Awarded ERGO Scholarship.

University of Florence

Florence, Italy

BSc in Computer Engineering — 110/110 cum laude — *Sep. 2019 - Apr. 2023*

- **Thesis**: Anomaly detection in intracardiac electrograms (EGMs) with dictionary learning, supervised by Prof. Paolo Frasconi.
- **Software foundations**: algorithms and data structures, software engineering, databases, theoretical computer science (C, Java, SQL).
- Received the DSU Scholarship three times.

Experience

LetXbe AI

Paris, France (Station F)

AI Intern — Document AI — *Feb. 2026 - Jun. 2026*

- Benchmarked open-weight Vision-Language Models for Document VQA on complex documents: PDFs, scans, forms and tables.
- Improved the training, evaluation and benchmarking pipeline (data quality checks, ground-truth validation, error analysis, reproducibility).
- Measured the effect of prompt format, image resolution, resizing strategy and input representation on VLM performance, reaching +5–10% accuracy over the baseline setup.
- Investigated interpretability through attention maps and internal model signals to locate the visual evidence behind each answer, making VLM-based Document AI more transparent and reliable.

Projects

Open-Set Semantic Segmentation on StreetHazards

GitHub · PyTorch, Hugging Face, W&B — *2026*

- Segmentation for driving scenes that also flags unknown objects: DINOv2 + ViT-Adapter backbone (62.98 mIoU) with likelihood-ratio OOD heads (ULRE, UEM) that raise anomaly AUPR from 17.2% to 50.9% and cut FPR95 from 19.2% to 11.0% vs the energy-score baseline.

SoccerAI — Shot Prediction with Graph Neural Networks

GitHub · PyTorch Geometric, Polars, W&B — *2025*

- Team of three. Predicts whether a possession ends in a shot from World Cup 2022 tracking and event data as per-frame player graphs (with scraped Transfermarkt/FBref stats); benchmarked 7 GNN backbones (GATv2, GCNII, GraphGPS, …) with temporal necks and explainability.

BONITA — Synthetic Instruction Data for Italian Boolean QA

GitHub · Transformers, PEFT/QLoRA, TRL — *2026*

- QLoRA fine-tune of Minerva-7B that turns unannotated Italian text into yes/no QA data (9,964 tasks from 10k Wikipedia paragraphs). A student trained only on this data reaches 88.0% accuracy vs 80.1% for the base model (+7.9 pp) on 2,490 human-labeled questions, on a single RTX 3090.

Hiring CV Bias — Auditing an Industrial CV Parser with Adecco

Project page · LLMs, Fairness Metrics, Streamlit — *2025*

- Team of two, on real anonymized CVs. Built an independent skill extractor (translation, LLM labelling, regex, exact and semantic ESCO matching) to audit a proprietary parser: recall of only 41–66% across skill types; fairness metrics (TPR parity, disparate impact) flag adverse impact on short CVs (DI 0.75).

Honors & Awards

2019 **Winner,** School Chess Team National Championship (Juniores) — *Montesilvano, Italy*

2018 **National Final Award, Best Business Plan,** JA Impresa in Azione — Junior Achievement Italy — *Milan, Italy*